

EXPERIMENT 2

1. AIM:

To design and implement an end-to-end ETL (Extract, Transform, Load) pipeline using Microsoft Power BI's Power Query Editor, enabling the extraction of data from a real-world business dataset, application of transformation logic, and loading of cleansed data into a unified analytical model for reporting and decision-making.

2. OBJECTIVES:

- To extract data from real-world business datasets using Microsoft Power BI.
- To transform and clean the extracted data using Power Query Editor.
- To apply data transformation techniques such as filtering, merging, removing duplicates, handling missing values, and changing data types
- To load the transformed data into a unified analytical model.

3. TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Sample datasets: [Global Superstore 2018.xlsx](#), Employee_DB.xlsx, and a sample SQL database

4. DATASET USED: [Global Superstore 2018.xlsx](#), Employee_DB.xlsx, Transactions.sql

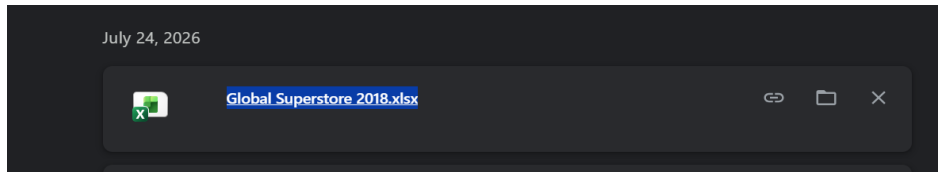
5. THEORY:

- ETL (Extract, Transform, Load) is one of the most important processes in Business Intelligence. It enables organizations to collect data from source systems, transform it into a clean and consistent format, and load it into a centralized environment for analysis.
- Extract Phase:
The extraction phase involves collecting data from source systems. In this experiment, data is extracted from the Orders, Returns, and People worksheets available in the Global Superstore dataset.
- Transform Phase:
The transformation phase includes data cleaning, handling missing values, correcting data types, creating derived columns, filtering invalid records, and integrating multiple datasets.
- Load Phase:
The transformed data is loaded into the Power BI data model where relationships can be established and analytical measures can be created using DAX.
- Power Query Editor serves as the ETL engine in Power BI and records each transformation step as an Applied Step, making the workflow transparent and reproducible.

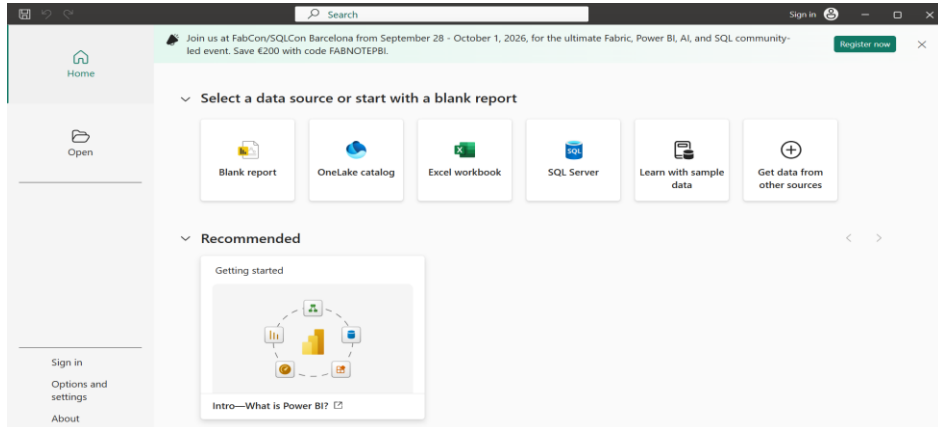
6. PROCEDURE:

Part A: Extract-Connect to Dataset

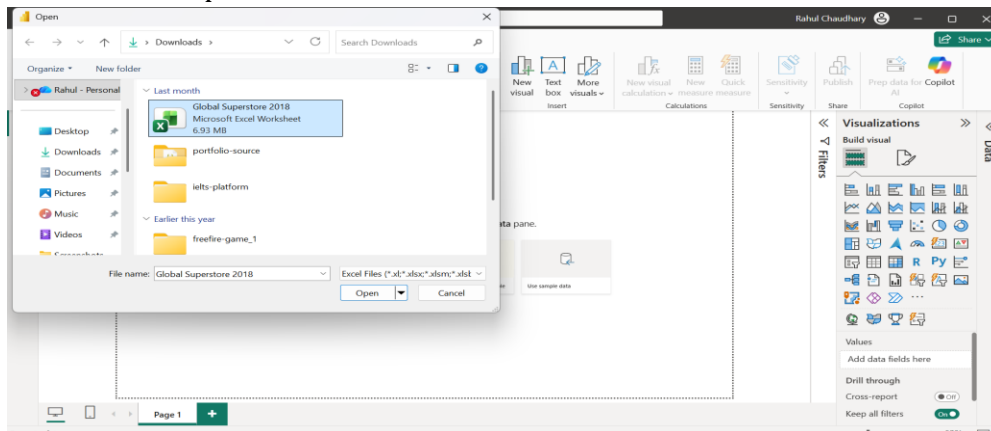
Step 1: Download the Global Superstore Dataset from Kaggle.



Step 2: Open Power BI Desktop.



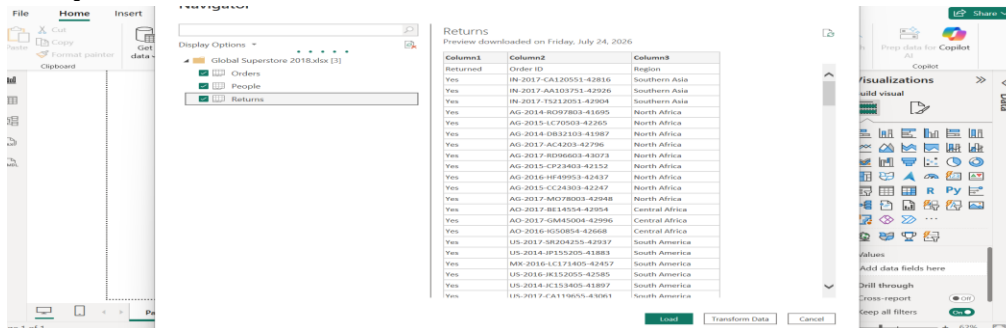
Home > Get Data > Excel Workbook
Select Global Superstore Dataset.xlsx



Click Transform Data.

Step 3: Import the following worksheets:

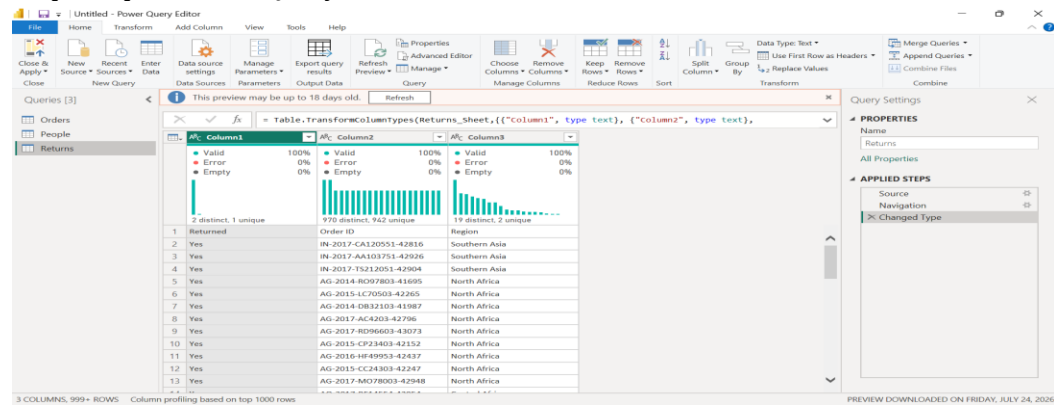
- Orders
- Returns
- People



Verify that all imported tables appear in the Queries pane.

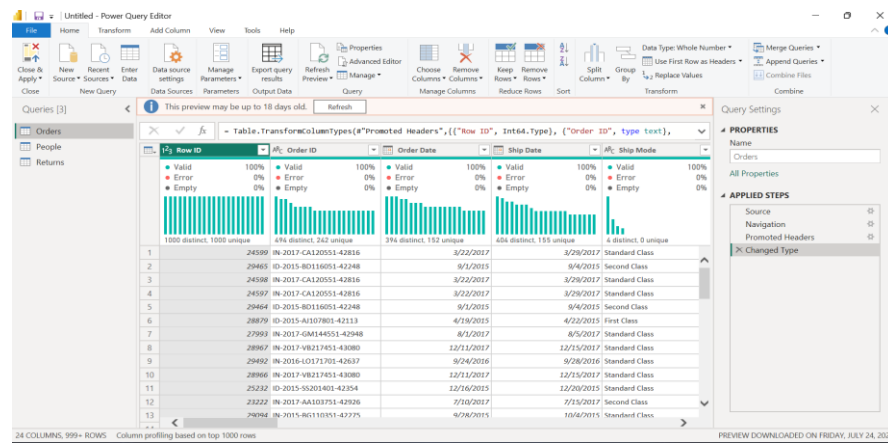
Part B: Transform-Power Query Editor

Step 4: Open Power Query Editor.

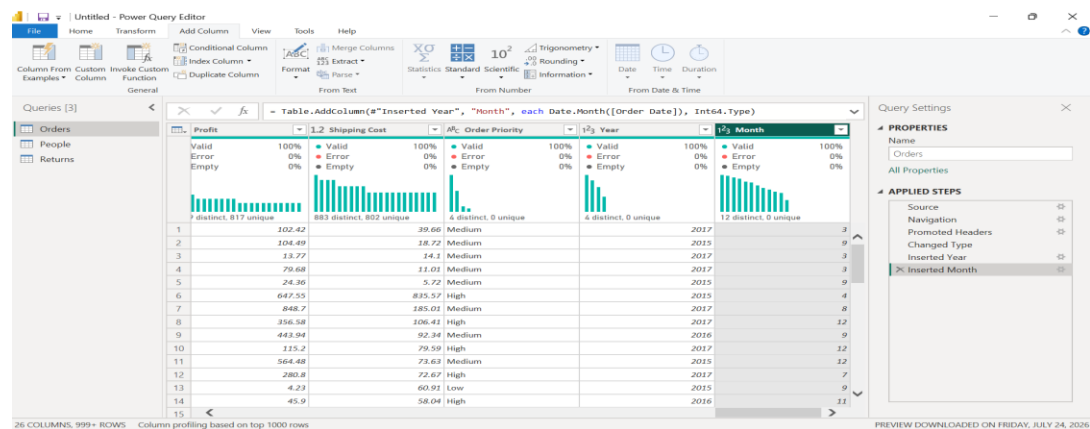


Step 5: Apply the following transformations to the Orders table:

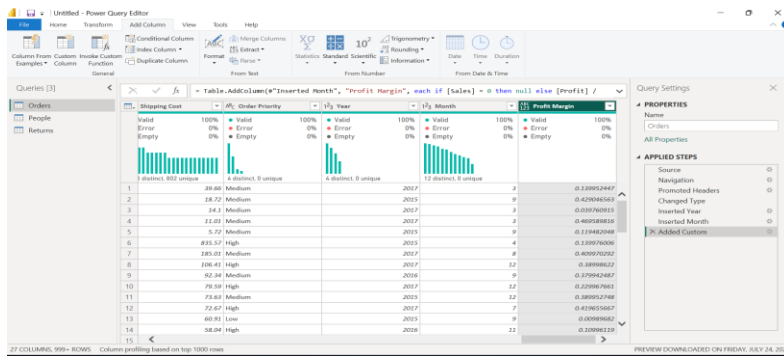
- Remove duplicate records
- Remove blank rows



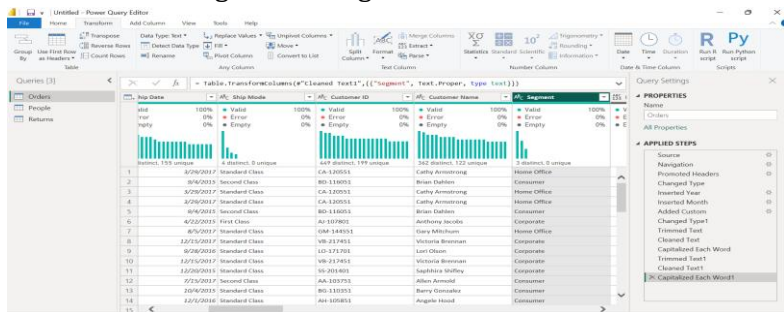
- Verify and correct data types
- Rename columns where required
- Extract Year and Month from Order Date



- Create a custom Profit Margin column

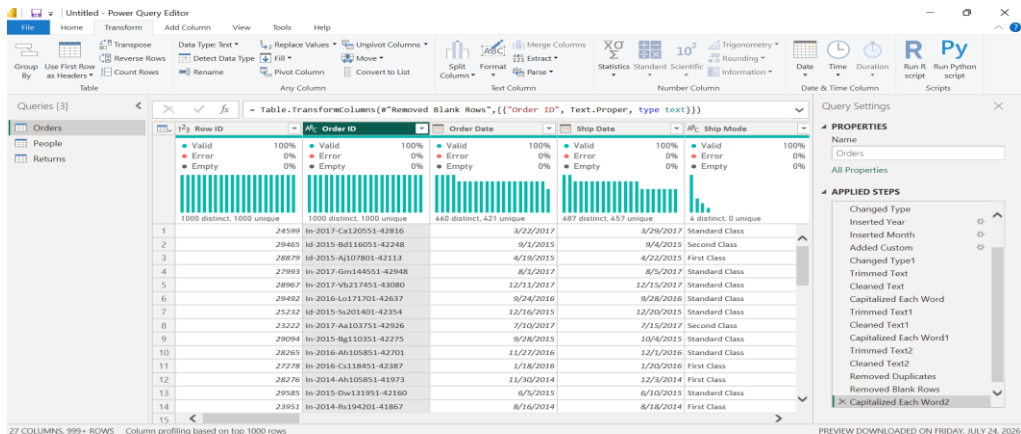


• Standardize Region and Segment values



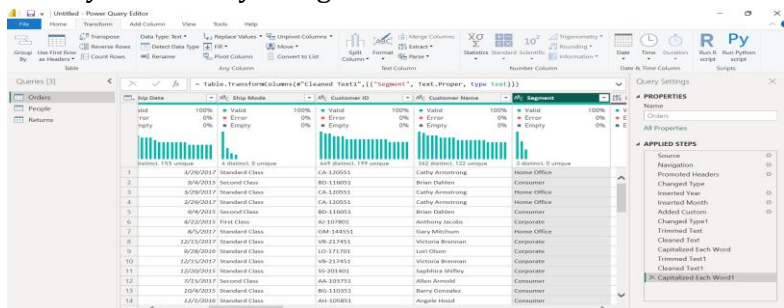
Step 6: Transform the Returns table:

- Remove duplicate Order IDs
- Check for missing values
- Standardize Returned field values

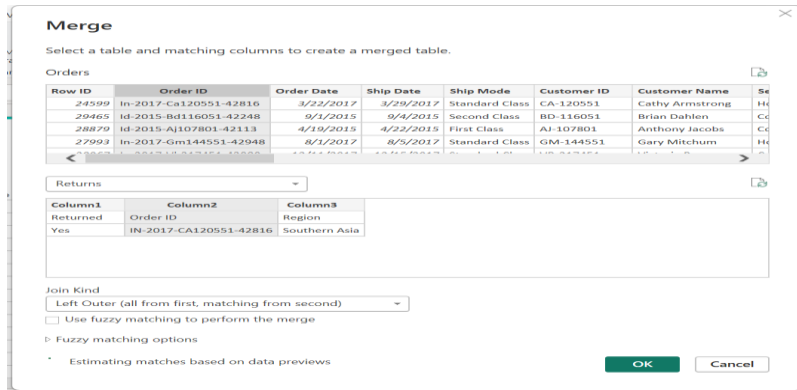


Step 7: Transform the People table:

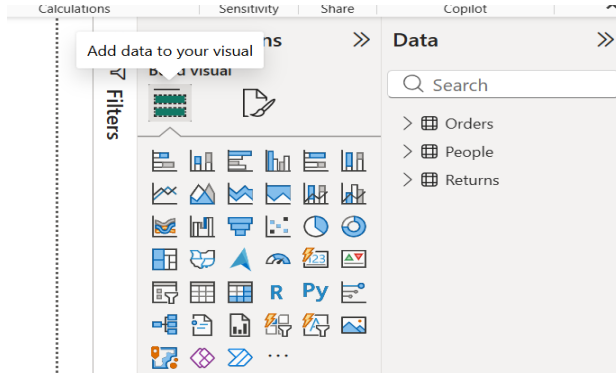
- Validate regional assignments
- Remove unnecessary columns
- Verify consistency of region names



Step 8: Merge Orders and Returns tables using Order ID.

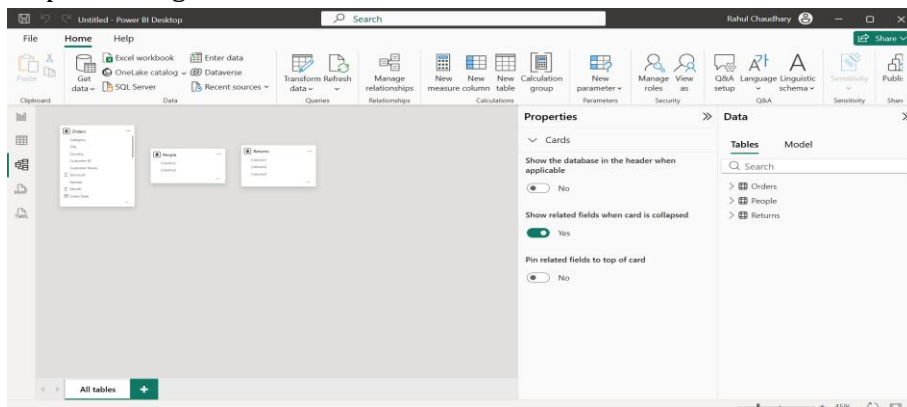


Step 9: Review all Applied Steps and click Close & Apply.



Part C: Load-Build Data Model

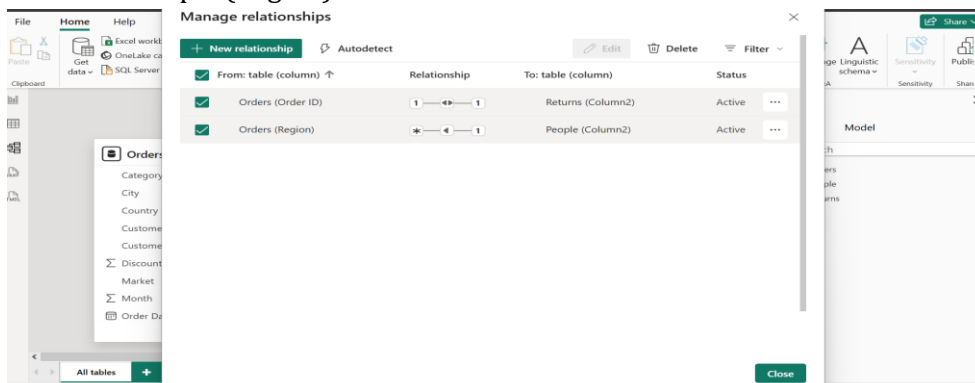
Step 10: Navigate to Model View.



Step 11: Create relationships:

Orders ↔ Returns (Order ID)

Orders ↔ People (Region)



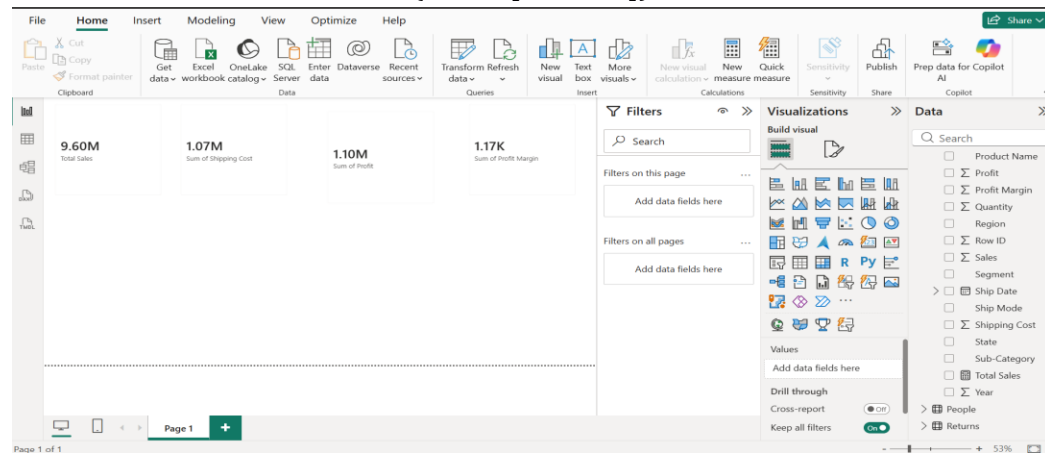
Step 12: Create DAX measures:

Total Sales = SUM(Orders[Sales])

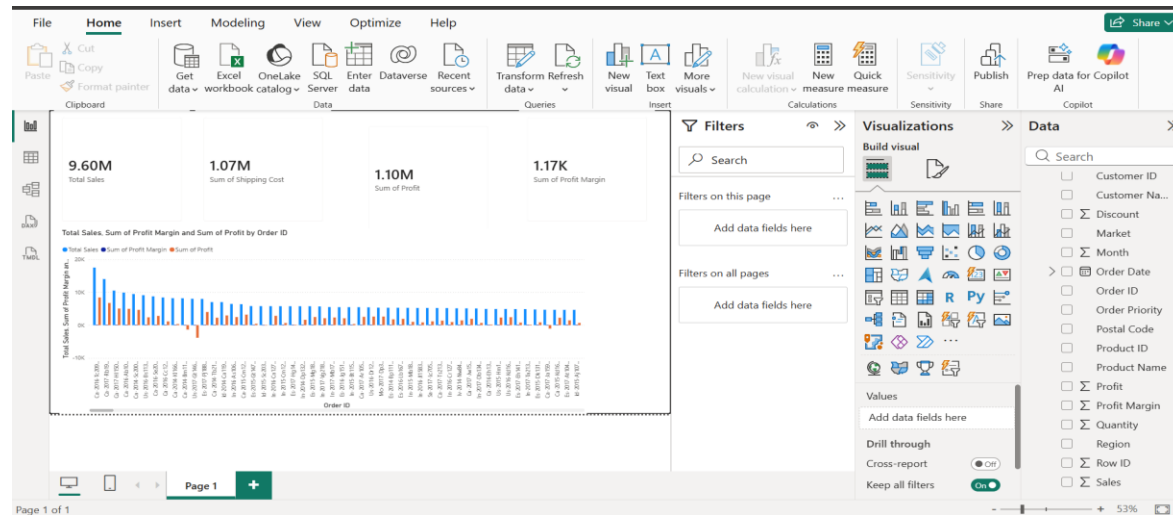
Total Profit = SUM(Orders[Profit])

Profit Margin = DIVIDE(SUM(Orders[Profit]),SUM(Orders[Sales]))

Total Orders = DISTINCTCOUNT(Orders[Order ID])



Step 13: Create a simple report visual using fields from related tables.



7. RESULT

Data was successfully extracted from the Global Superstore dataset, transformed using Power Query Editor, and loaded into the Power BI data model. Data cleaning, standardization, relationship creation, and DAX-based metric generation were performed successfully. The final dataset was prepared for reporting and business analysis.

8. LEARNING OUTCOMES

After completing this experiment, students will be able to:

- Explain the ETL process and its role in Business Intelligence systems. (CO1-BT2)
- Extract data from real-world business datasets using Power BI. (CO2-BT3)
- Apply transformation techniques including cleaning, filtering, and merging. (CO2-BT3)
- Create relationships and analytical measures in Power BI. (CO3-BT4)
- Validate transformed datasets for reporting and decision-making. (CO3-BT4)